TEAM MOMS · TRACK 2 · GAGNAÞON 2026

RAGstofan

Ask Iceland’s official statistics in plain Icelandic or English. Get the number, the chart and the table it came from.

Hagstofa Íslands · 2,000+ tables · one DGX Spark · open-source models only

The numbers exist. Finding them is the hard part.

2,000+

tables in the PX-API

Each with its own dimensions, codes, bases and periods. The right one is rarely the first search hit.

2

languages, one question

People ask in Icelandic or English, with everyday words: “verðbólga”, “hvað búa margir í Kópavogi”.

0

tolerance for made-up numbers

A language model alone will guess a plausible figure. Official statistics can’t.

ONE ANSWER

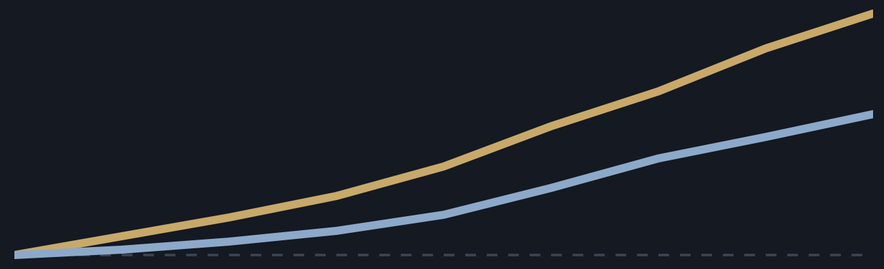
A sentence, the figures, a chart and the source.

“Hvernig hefur launavísitala og neysluverðsvísitala þróast síðustu 10 ár?”

Both series on one chart, each set to 100 in January 2016: wages at **211**, prices at **162**. Wages grew faster than prices.

1.149,3
Launavísitala · ágú. 2026
+110,9%

697,3
Vísitala neysluverðs · sep. 2026
+62,8%



Heimild · Hagstofa Íslands · LAU04000, VIS01000

A harness around the model

1 · Understand
Follow-ups made whole; facts kept only if the user said them

2 · Find
One table first, each candidate confirmed yes or no

3 · Split
Only if no table fits: 2, 3, then 4 parts

4 · Read
Categories chosen by the model, checked by code

5 · Calculate
Python, never the model

6 · Answer
Chart by the data’s shape, every number checked

We did not rely on a large model knowing the answer. The model makes small, checkable decisions with a fixed output schema; code checks each choice against the table, does the arithmetic and has the last word on every number.

Finding the table was our main focus

With the wrong table, no model gives the right answer. We look the way a person would, from the most reliable way to the broadest.

Every candidate gets one question

“Can this table answer the question, yes or no?” None passes: split the question. Still none: the user picks from the best candidates.

1 · The dictionary

212 subjects, 2,810 Icelandic and English words. verðbólga → consumer price index → VIS01000.

2 · Search

BM25 for exact words, e5 vectors for meaning across languages; the two rankings merged.

3 · The model reads the list

All table titles, sized to its context window; it names its best three.

NUMBERS YOU CAN TRUST

Every number in the answer is checked against the table before you see it.

Arithmetic is done in Python. If the model’s sentence contains a number the table doesn’t back, it is thrown away and the facts are stated plainly instead.

Test sets we built to measure table finding

60

Organisers’ questions

Easy, medium and hard, in English and Icelandic, with official answers.

~1,300

Written from 141 tables

Lookups, changes, categories, rewordings and near-identical table pairs. Half frozen for measuring only.

108

From Hagstofan’s news

Real topics and phrasing; kept only if the answer appears word for word in the release.

No answer keys written by hand: the table a question was written from is the right answer, and the news questions use Statistics Iceland’s own figures.

Results on the organisers’ questions

9/10

Right table, final system

10 questions drawn at random from the 60

8/10

Right answer, final system

The number matches the official one

59/60

Earlier fast version, all 60

Qwen3.6 35B-A3B · 7.4 s median · 5 model calls

Search alone, no model: the right table was in the top 15 for 78 of 90 questions with BM25 and e5 merged, against 74 for BM25 and 62 for e5.

The table finder was redesigned late, so the final system has one 10-question run. Earlier versions were tuned on the same 60, so they are not an unseen test.

A CHOICE WE MADE

We had a fast version. We chose a slower, more robust one.

Before: fast

Working and quick, about 7 seconds per answer. But the model chose among all candidate tables at once, with too little grounding and too few checks around it. It scored well on questions it was tuned on.

Now: harnessed

Every table is confirmed against the question, the question is split only when needed, and code checks each step. More model calls, so slower, but grounded in the data at every step.

How small can the model be?

Model (open weights)	Size	Right, of 20	EN · IS	Median
Qwen3.6 35B-A3B (3B active)	22 GB	19	10 · 9	41 s
Qwen3.8 27B	17 GB	17	10 · 7	53 s
gpt-oss 20B	13 GB	14	9 · 5	43 s
Qwen3 14B	9 GB	14	8 · 6	36 s
Qwen3 8B	5 GB	12	8 · 4	40 s
Qwen3 4B	2.5 GB	9	5 · 4	25 s

A mixture-of-experts model with 3B active parameters was both the most accurate and the fastest. Smaller dense models keep English longest; Icelandic is the first to break.

Earlier version of the system · the 20 hard questions of the organisers’ set, same data for every model, thinking off.

One engine, four ways in

Website

Live steps, charts, saved chats, private mode

iPad & iPhone app

Native SwiftUI, working today on iPad: live steps, charts, voice, Live Activity on the lock screen

Voice

Ask aloud and hear the answer, in Icelandic or English

API & MCP

The Track 2 endpoint, plus an MCP server for Claude and ChatGPT

Runs on one box, nothing leaves it

An NVIDIA DGX Spark runs the language model, search, speech and the server. Every model is open source and free.

Language model	Qwen3.6 35B-A3B via Ollama
Search	BM25 + e5 vectors
Speech in	Whisper large-v3-turbo
Speech out	Piper (Steinn, Ugla), Kokoro
Data	Local snapshot, live PX-API fallback
Access	Tailscale, starts at boot

LIVE DEMO · IPAD APP

Live, on our iPad app, from the Spark

Nothing recorded: the native iPad app asks the DGX Spark in real time.

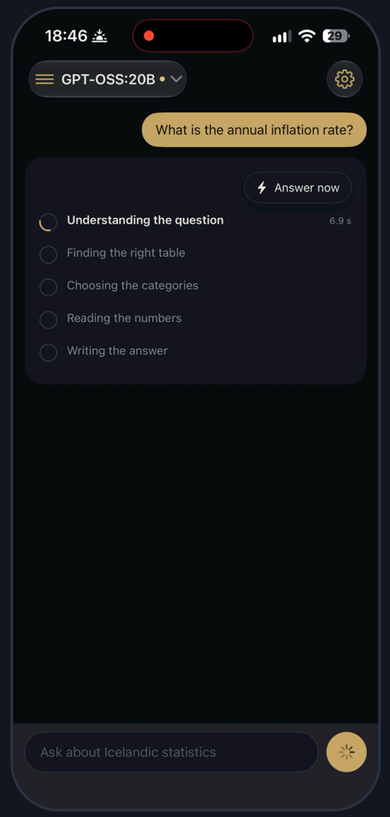
1. “Hve margir búa á Íslandi?” — one number, one table, by voice
2. “How have wages and prices developed over the last 10 years?” — two tables, one chart
3. “Hverjir eru helstu áhrifaþættir neysluvísitölunnar?” — ranked, then open a category
4. Lock the iPad mid-question — the answer arrives on the lock screen

IPHONE & IPAD APP

The native app, in use

Recorded on an iPhone: each step streams in live, the answer comes with figures and a chart, and the lock screen shows progress while you wait.

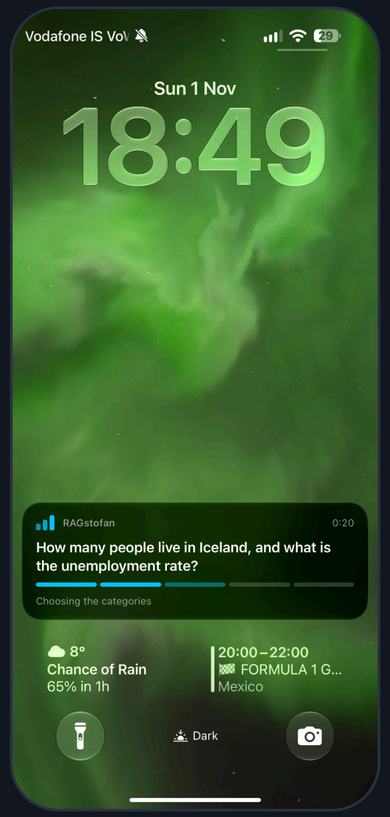
The same app runs on iPad. That’s the one in the live demo.



Live steps



Answer and chart



Lock screen

Where it falls short

The table is usually right. The misses are in the calculation after it.

- Dividing across two tables (per inhabitant)
- Summing several categories
- A total over all ages or countries when the table has none

What we'd do next

- Decide more reliably which calculation is asked for
- Check units before combining: no summing percentages
- A test set aimed at calculations, as we built for tables
- Pick the model by difficulty: small and fast for lookups, the strongest for hard questions

Built with AI coding agents

Claude Code and OpenAI Codex wrote and changed code, ran tests and evaluations, and wrote the docs. The team set the direction, decided the design and reviewed the work.

94/105

commits on main co-written with Claude Code; Codex work on **codex/** branches

Takk fyrir.

Questions? Ask us, or ask RAGstofan.

Team MOMS · [team members' names] · [link to the app or website]

